

Algebra I — Lecture 7

Subgroups and Cosets

Lecture notes

1 Subgroups

Throughout, G denotes a group with identity e .

Definition 1.1. A nonempty subset $H \subseteq G$ is a *subgroup*, written $H \leq G$, if $ab^{-1} \in H$ for all $a, b \in H$.

Example 1.2. For every $n \geq 1$ the multiples $n\mathbb{Z}$ form a subgroup of $(\mathbb{Z}, +)$.

Proposition 1.3. *The intersection of any family of subgroups of G is a subgroup of G .*

Proof. Each subgroup contains e , so the intersection is nonempty. If a and b lie in every member of the family, so does ab^{-1} . $\square$

Remark. The union of two subgroups is rarely a subgroup: $2\mathbb{Z} \cup 3\mathbb{Z}$ contains 2 and 3 but not 5.

2 Cosets and Lagrange's theorem

Definition 2.1. For $H \leq G$ and $g \in G$, the *left coset* of H containing g is $gH = \{gh : h \in H\}$. The number of left cosets is the *index* $[G : H]$.

Lemma 2.2. *The left cosets of H partition G , and any two of them have the same cardinality.*

Proof. The relation $a \sim b \iff a^{-1}b \in H$ is an equivalence relation whose classes are exactly the left cosets. The map $h \mapsto gh$ is a bijection from H onto gH . $\square$

Theorem 2.3 (Lagrange). *If G is finite and $H \leq G$, then $|G| = [G : H] |H|$. In particular $|H|$ divides $|G|$.*

Proof. By Lemma 2.2, G is the disjoint union of $[G : H]$ cosets, each with $|H|$ elements. $\square$

Corollary 2.4. *For every g in a finite group G , $\text{ord}(g)$ divides $|G|$, and $g^{|G|} = e$.*

Proof. Apply Theorem 2.3 to the cyclic subgroup $\langle g \rangle$, whose order is $\text{ord}(g)$. $\square$

Corollary 2.5 (Fermat). *If p is prime and $p \nmid a$, then $a^{p-1} \equiv 1 \pmod{p}$.*

Example 2.6. The group S_3 has order 6, so its subgroups have order 1, 2, 3 or 6. Every one of these orders occurs.